

BUECKEBURG, Germany

WMO: 103350

Lat: **52.2817N**Lon: **9.0889E**Elev: **251**StdP: **14.56**Time Zone: **1.00 (EUC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.9	20.7	10.3	9.4	19.3	15.2	12.0	22.5	29.9	45.7	27.3	45.8	6.1	80	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.9	87.5	67.7	83.4	66.4	79.5	65.0	69.6	82.9	67.8	79.9	66.2	77.1	8.0	170

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.2	93.9	73.8	63.7	89.1	72.2	62.2	84.5	70.9	33.7	83.1	32.2	80.0	31.0	77.2	80.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.6	20.8	18.4	11.2	93.6	7.3	3.0	5.9	95.8	1.6	97.5	-2.6	99.1	-7.9	101.3
			10.6	73.3	7.2	2.4	5.4	75.0	1.2	76.4	-2.8	77.8	-8.0	79.5
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	50.9	36.2	37.8	42.3	49.7	56.5	62.1	66.3	66.0	59.3	51.7	43.7	38.3
	DBStd	12.71	8.47	7.99	7.03	7.09	6.60	6.17	6.08	5.77	5.68	6.60	6.71	7.95
	HDD50	1770	431	345	252	93	15	0	0	0	1	59	207	367
	HDD65	5405	894	762	704	460	274	129	57	56	189	412	640	828
	CDD50	2095	1	3	13	83	218	363	504	495	281	113	17	4
	CDD65	255	0	0	0	1	12	41	96	86	18	1	0	0
	CDH74	2370	0	0	0	14	123	393	925	775	136	4	0	0
	CDH80	737	0	0	0	0	20	101	321	267	28	0	0	0
	WSAvg	8.3	10.4	9.8	9.3	7.7	7.2	7.0	6.9	6.9	7.4	8.4	8.9	9.9
	PrecAvg	29.4	2.6	2.0	1.9	1.8	2.4	2.5	3.0	3.1	2.5	2.5	2.5	2.6
(15) Precipitation	PrecMax	40.3	4.7	4.3	4.3	3.7	5.9	5.4	7.1	8.0	5.6	7.4	5.0	6.3
	PrecMin	21.6	0.4	0.6	0.3	0.2	0.9	0.9	1.1	0.7	0.6	0.6	0.2	0.8
	PrecStd	4.6	1.2	1.0	1.1	0.9	1.3	1.0	1.3	1.5	1.2	1.3	1.2	1.1
	PrecStd	4.6	1.2	1.0	1.1	0.9	1.3	1.0	1.3	1.5	1.2	1.3	1.2	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	55.3	57.7	66.1	75.9	83.0	88.1	92.6	91.8	84.2	72.0	61.1	56.4	
	0.4% MCWB	49.3	48.5	52.2	59.1	65.6	69.6	68.5	68.8	65.9	61.9	54.9	51.9	
	2% DB	51.9	53.7	60.5	70.5	77.2	82.6	87.8	87.1	78.1	67.0	57.5	53.8	
	2% MCWB	48.0	47.2	50.7	56.6	63.2	67.0	67.7	67.2	64.2	59.2	52.8	50.0	
	5% DB	49.1	50.9	55.7	66.6	73.3	78.6	83.8	82.2	73.1	64.0	55.4	51.5	
	5% MCWB	45.9	46.3	48.7	54.6	60.2	64.9	66.4	66.0	62.0	57.8	51.6	48.0	
	10% DB	46.6	48.4	52.6	62.7	69.6	74.5	79.1	77.4	69.4	61.0	53.0	48.6	
	10% MCWB	43.5	44.4	46.7	53.2	58.4	62.7	65.1	64.1	60.2	56.3	49.3	45.6	
	0.4% WB	51.1	50.4	54.8	60.4	68.5	71.5	72.0	71.2	67.7	62.6	56.6	52.8	
	0.4% MCDB	54.0	54.1	62.5	72.2	79.8	84.5	85.4	84.3	79.3	70.5	59.7	55.6	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	48.5	48.6	51.8	58.0	64.3	68.3	69.6	69.0	65.4	60.1	53.9	50.7	
	2% MCDB	51.6	52.2	57.8	68.2	74.8	79.4	83.2	82.7	75.7	65.8	57.1	53.4	
	5% WB	46.0	46.5	49.5	55.9	61.4	65.9	67.8	67.1	63.2	58.5	51.7	48.3	
	5% MCDB	48.8	50.4	54.9	64.8	71.0	76.6	80.1	79.0	71.3	63.2	55.3	51.1	
	10% WB	43.6	44.4	47.2	53.7	59.0	63.7	66.0	65.4	61.2	56.6	49.6	45.8	
	10% MCDB	46.1	48.2	52.1	61.6	67.6	72.3	77.1	75.3	67.8	60.7	52.4	48.3	
(35) Mean Daily Temperature Range	MDBR	7.9	9.9	13.0	17.1	17.7	17.8	17.9	17.6	15.2	12.0	8.4	7.6	
	5% DB	9.6	12.2	19.0	24.0	24.2	24.8	26.0	26.0	21.4	14.9	10.3	8.4	
	MCDBR	7.9	8.4	11.9	13.2	13.0	12.5	10.8	11.2	11.0	9.2	7.7	7.2	
	5% WB	9.1	10.4	16.6	21.2	21.7	22.2	23.2	21.6	18.7	13.6	10.1	8.7	
	MCWBR	8.1	8.1	11.1	12.4	12.4	12.1	10.3	10.4	10.4	9.0	8.1	7.9	
	MCWBR	8.1	8.1	11.1	12.4	12.4	12.1	10.3	10.4	10.4	9.0	8.1	7.9	
(40) Clear-Sky Solar Irradiance	taub	0.320	0.342	0.377	0.412	0.414	0.413	0.427	0.420	0.389	0.358	0.339	0.316	
	taud	2.360	2.342	2.263	2.208	2.239	2.262	2.234	2.270	2.348	2.414	2.384	2.353	
	Ebn at Noon	213	243	255	260	266	267	260	255	250	234	203	192	
	Edh at Noon	19	26	34	41	41	41	41	38	31	24	19	16	
	RadAvg	204	422	760	1248	1522	1632	1560	1313	955	550	247	159	
(45) All-Sky Solar Radiation	RadStd	23	62	90	150	163	129	207	104	121	53	30	16	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d) +0.63	(e) N/A	(f) +3.20	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) -207	(l) N/A	(m) N/A
Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	-218	N/A	N/A